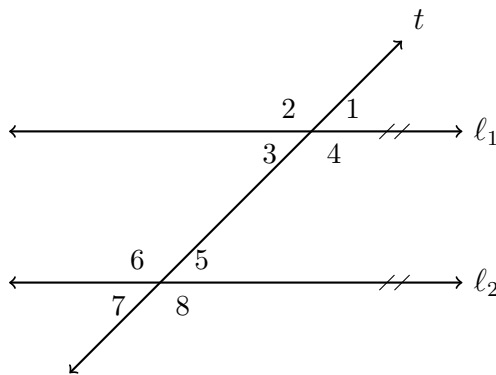


Skills Summary

Angle Pairs Formed by a Transversal

Use this reference while completing your worksheet or when studying.



Interior angles (3, 4, 5, 6) lie between the parallel lines; **exterior** angles (1, 2, 7, 8) lie outside them.

Skill 1 — Naming the pair before you compute

Two questions name every pair: inside or outside the parallel lines, and same side of t or opposite sides?

- **Corresponding** — same corner at the two intersections (1 and 5). *Congruent.*
- **Alternate interior** — inside, opposite sides (4 and 6). *Congruent.*
- **Alternate exterior** — outside, opposite sides (2 and 8). *Congruent.*
- **Same-side interior** — inside, same side (4 and 5). *Supplementary.*

Example: $m\angle 2 = 130^\circ$. Find $m\angle 6$.

Step 1. Angles 2 and 6 sit in the same corner at their own intersections, so they are corresponding — and corresponding angles on parallel lines are congruent, not supplementary.

Step 2. $m\angle 6 = m\angle 2 = 130^\circ$

Try it: $m\angle 3 = 47^\circ$. Find $m\angle 5$. check: 47°

Skill 2 — Finding a missing measure

Rule: Congruent pairs (corresponding, alternate interior, alternate exterior) copy the measure. Supplementary pairs (linear pairs, same-side interior) subtract it from 180° . Chain the two: name a pair, use it, then name the next.

Example: $m\angle 1 = 118^\circ$. Find $m\angle 4$.

Step 1. Angles 1 and 4 are adjacent at the same intersection — a linear pair — so this is a subtraction, not a copy.

Step 2. $m\angle 4 = 180 - 118 = 62^\circ$

Try it: $m\angle 4 = 84^\circ$. Find $m\angle 5$ (same-side interior). check: 96

Skill 3 — Solving for x from an angle pair

Rule: The pair name chooses the equation: congruent pair $\Rightarrow$ set the two expressions **equal**; supplementary pair $\Rightarrow$ set their **sum** to 180. Solve, then substitute back to get the measure the question actually asked for.

Example: Alternate interior angles measure $(5x - 12)^\circ$ and $(3x + 20)^\circ$. Find x .

Step 1. Alternate interior angles are congruent, so the two expressions are equal — no 180 appears anywhere in this setup.

Step 2. $5x - 12 = 3x + 20 \Rightarrow 2x = 32 \Rightarrow x = 16$

Try it: Same-side interior angles measure $(4x + 10)^\circ$ and $(6x)^\circ$. Find x . check: $x = 11$